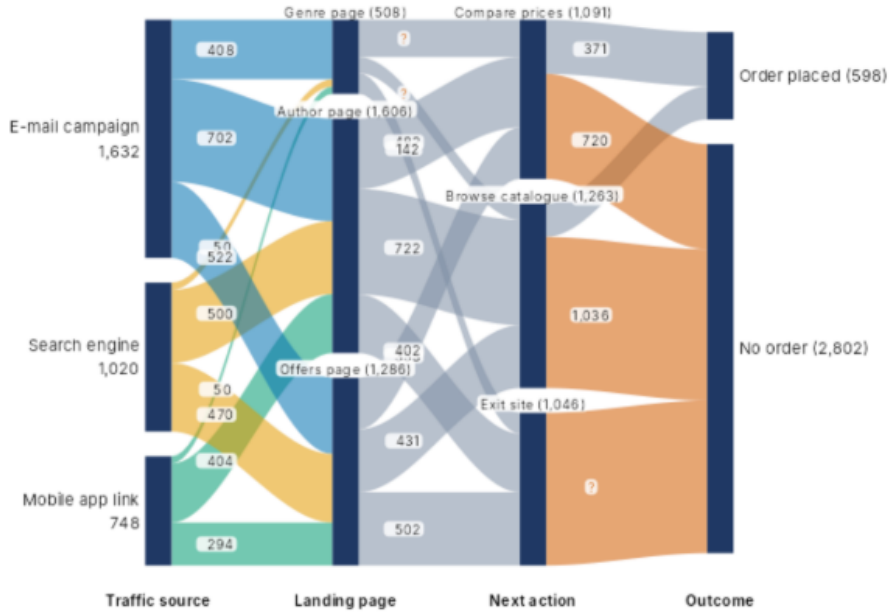
Question

DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot D

R.No: 12 Reg. No: 192424277 Name: GUNA P

Scenario: The analytics team of an online bookstore tracked 3,400 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Exit site to No order; Genre page to Browse catalogue; Genre page to Compare prices. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Order placed as a percentage of all sessions; (ii) the largest single flow into "No order" or "Exit site"; (iii) the percentage of "Offers page" sessions that go on to "Compare prices". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 200 sessions. Compute the height in cm of each landing-page node (Author page, Offers page, Genre page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Search engine" and "Mobile app link" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

DSA06 - Data Handling and Visualization

Activity 6: Website Flow Sankey Design | Slot D

Name: GUNA P | R.No: 12 | Reg. No: 192424277

(a) Hidden flows by conservation of flow [4]

Rule used: for every node, sum of incoming flows = node total = sum of outgoing flows.

1. Exit site to No order. Order placed (598) receives only Compare prices to Order 371 and Browse catalogue to Order. Browse catalogue = 1,263 and Browse to No order = 1,036, so Browse to Order = 1,263 - 1,036 = 227. Then 371 + 227 = 598, so Exit site sends 0 to Order placed. All 1,046 Exit-site sessions go to No order.
Check on No order: 720 + 1,036 + 1,046 = 2,802.

Exit site to No order = 1,046

2. Genre page to Compare prices. Exit site (1,046) = Genre 142 + Author 402 + Offers 502 = 1,046, so Genre to Exit = 142. Genre page (508): Compare + Browse = 508 - 142 = 366. Compare prices (1,091) = Genre + Author 482 + Offers 353, so Genre to Compare = 1,091 - 482 - 353 = 256.

Genre page to Compare prices = 256

3. Genre page to Browse catalogue. 366 - 256 = 110. Check on Browse catalogue: 110 + 722 + 431 = 1,263.

Genre page to Browse catalogue = 110

Complete flow table (used in the later parts):

From	To	Sessions
E-mail campaign	Genre / Author / Offers	408 / 702 / 522
Search engine	Genre / Author / Offers	50 / 500 / 470
Mobile app link	Genre / Author / Offers	50 / 404 / 294
Genre page (508)	Compare / Browse / Exit	256 / 110 / 142
Author page (1,606)	Compare / Browse / Exit	482 / 722 / 402
Offers page (1,286)	Compare / Browse / Exit	353 / 431 / 502
Compare prices (1,091)	Order / No order	371 / 720
Browse catalogue (1,263)	Order / No order	227 / 1,036
Exit site (1,046)	Order / No order	0 / 1,046

(b) Computations [4]

(i) Overall conversion rate = 598 / 3,400 = 0.1759 = **17.6%**

(ii) Largest single flow into No order or Exit site. Flows into No order: 720, 1,036, 1,046. Flows into Exit site: 142, 402, 502. The largest is **Exit site to No order = 1,046 sessions**, which is 1,046 / 3,400 = 30.8% of all sessions (just ahead of Browse to No order, 1,036).

(iii) Offers page sessions going to Compare prices = 353 / 1,286 = 0.2745 = **27.4%**

(c) Four design decisions [4]

Decision	Choice	One-line justification
Node order	Order each stage so that the most-converting node is on top: Compare prices, then Browse catalogue, then Exit site; Order placed above No order.	Conversion is 371/1,091 = 34.0% for Compare, 227/1,263 = 18.0% for Browse and 0% for Exit, so the Order flows stay short and almost nothing crosses.
Colour rule	Colour each flow by traffic source (3 hues); use one neutral grey for Exit site to No order.	The reader can trace where E-mail (1,632), Search (1,020) and Mobile (748) traffic ends up; grey marks the drop-off.
Labels	Label every node with its count and % of 3,400 (e.g. Order placed 598 = 17.6%); label only flows of about 200 or more.	Small flows such as 50, 110 and 142 would clutter the diagram; the node totals already carry the key numbers.
What to highlight	Emphasise the Exit site to No order flow (1,046) and the Compare prices to Order flow (371) in strong colour; fade the rest.	Exit leakage is the largest loss (30.8% of sessions) and Compare is the best path to an order, so these two tell the story.

(d) Node heights at 1 cm = 200 sessions [2]

Landing node	Sessions	Height = sessions / 200
Author page	1,606	8.03 cm
Offers page	1,286	6.43 cm
Genre page	508	2.54 cm

Together: 8.03 + 6.43 + 2.54 = 17.0 cm = 3,400 / 200, so the column is consistent.

(e) R code (networkD3) [4]

The four flows leaving Search engine and Mobile app link, in diagram order from the top, are Search to Genre 50, Search to Author 500, Search to Offers 470 and Mobile to Genre 50. They are the first rows of the data frame, followed by all other flows so the whole diagram is drawn.

```
library(networkD3)

# ---- nodes (order within each stage = top-to-bottom order in the diagram)
nodes <- data.frame(name = c(
  "E-mail campaign", "Search engine", "Mobile app link",      # 0-2  Traffic source
  "Genre page", "Author page", "Offers page",                  # 3-5  Landing page
  "Compare prices", "Browse catalogue", "Exit site",           # 6-8  Next action
  "Order placed", "No order"))                                 # 9-10 Outcome

# ---- links: first four flows leaving Search engine / Mobile app link
links4 <- data.frame(
  source = c("Search engine", "Search engine", "Search engine", "Mobile app link"),
  target = c("Genre page",    "Author page",   "Offers page",   "Genre page"),
  value  = c(50, 500, 470, 50))

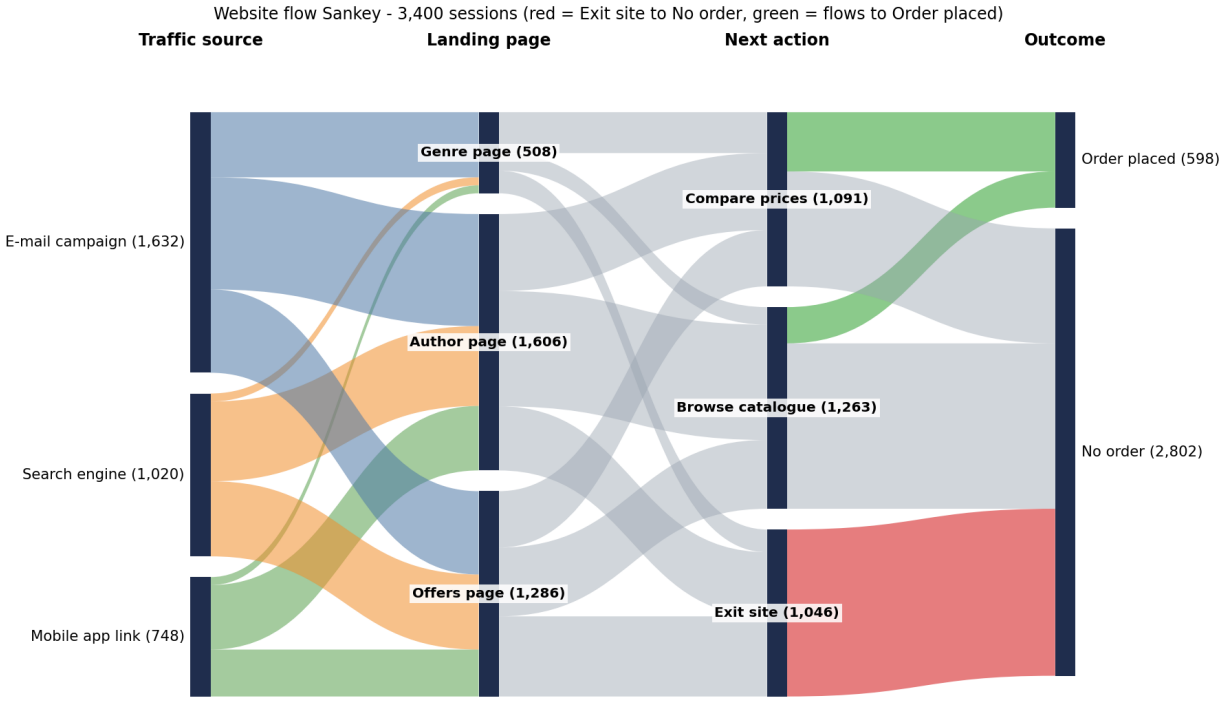
# ---- remaining flows
links_rest <- data.frame(
  source = c(rep("E-mail campaign", 3), rep("Mobile app link", 2),
             rep("Genre page", 3), rep("Author page", 3), rep("Offers page", 3),
             "Compare prices", "Compare prices", "Browse catalogue",
             "Browse catalogue", "Exit site"),
  target = c("Genre page", "Author page", "Offers page", "Author page", "Offers page",
             rep(c("Compare prices", "Browse catalogue", "Exit site"), 3),
             "Order placed", "No order", "Order placed", "No order", "No order"),
  value  = c(408, 702, 522, 404, 294,
             256, 110, 142,  482, 722, 402,  353, 431, 502,
             371, 720, 227, 1036, 1046))

links <- rbind(links4, links_rest)
stopifnot(sum(links$value[links$source %in% nodes$name[1:3]]) == 3400)

# networkD3 needs 0-based node indices
links$IDsource <- match(links$source, nodes$name) - 1
links$IDtarget <- match(links$target, nodes$name) - 1

sankeyNetwork(Links = links, Nodes = nodes,
              Source = "IDsource", Target = "IDtarget",
              Value = "value", NodeID = "name",
              units = "sessions", fontSize = 12, nodeWidth = 30,
              nodePadding = 12, iterations = 0,   # iterations = 0 keeps the node order above
              sinksRight = FALSE)
```

Output of the code (Sankey diagram, node order as in the nodes data frame):



In RStudio the sankeyNetwork() call shows this diagram in the Viewer pane; the image above is the same data and node order.

(f) When a Sankey misleads or becomes unreadable [2]

1. Too many nodes and flows (for example hundreds of individual pages): the ribbons tangle and cross, and nobody can follow one path. **Alternative:** a heat map (transition matrix) of from-page by to-page.

2. Comparing exact rates or many small flows (for example conversion by source, where flows of 50 and 110 become thin slivers, and ribbon width is hard to compare). **Alternative:** a bar chart of conversion rate per source or step. (A Sankey also cannot show loops or return visits; a network graph suits those.)